13 May 2026

Professor Neil Ghani

Hi Neil,

I've been reading through some recent work on multi-agent LLM systems and thought you might find it interesting, given our conversations about hierarchical composition and coordination. Three pieces stood out:

1. SciAgent: A Unified Multi-Agent System for Generalistic Scientific Reasoning
arXiv:2511.08151 — Note: currently withdrawn pending revision

SciAgent targets cross-disciplinary scientific reasoning at expert level. A *Coordinator Agent* classifies the domain and difficulty of each problem, then dynamically assembles a *Worker System* of sub-agents handling symbolic deduction, conceptual modelling, numerical computation, and verification. The claimed results were striking—gold-medalist level on the IMO, IPhO, IChO, and Humanity's Last Exam—but the paper has since been withdrawn by the authors pending further expert evaluation. I include it here as an interesting architecture sketch, not as a reliable empirical claim.

2. AgentOrchestra: A Hierarchical Multi-Agent Framework for General-Purpose Task Solving
arXiv:2506.12508

AgentOrchestra is a two-tier framework targeting general task-solving rather than a specific domain. A *Planning Agent* decomposes tasks and routes work to three specialists:

- *Deep Researcher* — broad web information gathering
- *Browser Use Agent* — precise web interaction and extraction
- *Deep Analyzer* — multimodal reasoning over retrieved content

The design principles (extensibility, multimodality, modularity, coordination) are deliberately architectural rather than empirical. Results on standard benchmarks:

Benchmark	AgentOrchestra	Previous SOTA
SimpleQA	95.3%	93.9% (Perplexity Deep Research)
GAIA	82.42%	77.58% (AWorld)
HLE	25.9%	20.3% (o3)

The HLE result in particular is interesting: 25.9% vs 20.3% for o3 is a meaningful jump on a benchmark designed to be extremely difficult.

3. Building a Multi-Agent Research System
Anthropic Engineering Blog, 2025

Anthropic describe their own internal multi-agent research assistant. The architecture is an orchestrator–worker pattern: a lead `claude-opus-4` agent builds a dynamic research strategy and spawns 3–5 parallel `claude-sonnet-4` subagents, each with its own context window, doing web search and extended thinking. A separate citation agent handles attribution post-hoc.

Key empirical findings:

- Multi-agent (Opus lead + Sonnet workers) outperforms single-agent Opus by **90.2%** on internal research evaluations
- Parallel execution cut research time by up to **90%** on complex queries
- Token usage explains **80%** of performance variance in browsing tasks
- Improved tool descriptions alone reduced task completion time by **40%**

The last point about token usage is the mechanistically interesting one: separate context windows per subagent function as parallel working memory. The performance gain is not purely about parallelism—it is about avoiding the degradation that comes from a single agent's context window filling up on a long multi-step task.

Cross-cutting observation

All three systems land on the same architectural form: a coordinator plus specialised workers, assembled hierarchically. What varies is what the coordinator *does*:

- *SciAgent*: classifies domain → selects reasoning modalities
- *AgentOrchestra*: decomposes task → routes to tool-appropriate specialists
- *Anthropic*: builds a dynamic research strategy → spawns parallel explorers without pre-committing to structure

There is something here that feels adjacent to what we have been thinking about with categorical composition: the coordinator is essentially choosing a *morphism* in some category of task-decompositions, and the specialists are the component objects. The hierarchical structure is not just engineering convenience—it is doing real semantic work in determining what kind of reasoning gets applied where.

I am curious whether you see a clean categorical account of the coordination problem. Seems like it could connect to your work on containers or polynomial functors.

Best,

Robin